Small Aerial Target Detection Using Trajectory Hypothesis and Verification

Liangchao Guo, Xiaoliang Sun, Wenlong Zhang, Zhang Li, and Qifeng Yu

***Abstract*— Due to cluttered backgrounds, long-range imaging, rapid relative motion, etc., the detection of small aerial targets remains a challenge. Existing works have illustrated the importance of temporal cues for robust small aerial target detection. However, the strong assumptions about the temporal cues give such methods high-computational complexity and weak adaptability. To address this problem, we treat trajectory verification as a classification problem using convolutional neural networks. A novel small aerial target detection method that uses trajectory hypothesis and verification is proposed in this study. First, the inertial measurement unit (IMU)-assisted interframe difference is used to extract target candidates from each single image. Then, based on the continuous and smoothness characteristics of the target trajectory, hypotheses of the required length are generated by linking target candidates in adjacent frames. Finally, the trajectory hypotheses are sent to a trained classification convolution neural network for verification. The detected targets are traced back from the verified trajectory. Furthermore, trajectory merge is conducted to link adjacent trajectory segments. The proposed method can achieve real-time processing. Experimental results on public datasets show that the proposed method performs better than existing methods.**

***Index Terms*— Infrared detection system, small aerial target, trajectory hypothesis, trajectory verification.**

I. Introduction

ROBUST and efficient infrared small aerial target detection against a cluttered background is the key technology in airborne situational awareness, flight guidance, etc. [1]. In addition, it has always been a hot topic in the field of infrared image processing. However, for infrared detection systems that are subject to rapid relative motion and cluttered backgrounds, small aerial target detection remains a challenge.

Infrared small target detection has been studied for decades. The information contained in a single frame or multiple successive frames is used for target detection. Accordingly, existing methods can be categorized into single-frame-based methods and multiple-successive-frames-based methods [1]. In a single infrared image, the differences between small targets and backgrounds are often used in target detection. The small infrared target is often modeled as a spot target of isotropic distribution [2], [3]. The distribution is determined based on the point spread function of the imaging system or is simply approximated by Gaussian distribution. The background characteristics [4], [5], [6] and local contrast [7], [8], [9] are also widely used in single-frame-based infrared small target detection. Although such methods are efficient and easy to implement, cues provided by a single frame may be inadequate for robust small infrared target detection, especially in cases where the targets are extremely weak and the backgrounds are cluttered. The method based on multiple frames introduces temporal cues through the association between consecutive frames to detect objects [10], [11]. Multiple-successive-frames-based methods boost the performance of small target detection by associating multiple images and have better potential and prospects. However, the adoption of temporal cues increases the computational complexity of existing methods. Additionally, such methods cannot handle rapidly changing backgrounds well. Recently, the method based on deep learning was shown to effectively boost the performance of small target detection through its powerful feature-learning ability [12], [13], [14]. However, such methods generally only utilize the spatial information of small targets and have poor generalization capability. In conclusion, existing methods, including single-frame-based, multiframe-based, and network-based methods, have difficulty detecting small airborne targets.

Temporal cues are of great significance for small target detection in image sequences. This study analyzes a large number of small aerial target image sequences. The backgrounds are varied, e.g., clouds, vegetation, water, and buildings. Cluttered backgrounds may not show high correlation within image sequences, especially for cases in which the background is changing rapidly due to the motion of the airborne infrared detection system. In this situation, the cues of the backgrounds are inappropriate choices for robust small infrared aerial target detection. However, we find that true targets exhibit a continuous and smooth long trajectory, whereas the clutter does not. Based on this fact, this study tackles the challenge of small aerial target detection using trajectory information. Focusing on the problem that existing methods based on trajectory have high-computational complexity and weak adaptability, this study treats the trajectory verification as a classification problem using convolutional neural networks and proposes a novel small target detection method using trajectory hypothesis and verification. Spatial and temporal cues within image sequences are used in a simple and effective way. Spatial cues from each frame are used in target candidate detection.

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The proposed method uses inertial measurement unit (IMU) to solve the homography matrix between adjacent frames. Then, the interframe difference method (IFDM) is adopted to detect the target candidates in each frame. Temporal cues are used in detecting the true targets from numerous target candidates. This study uses a distance constraint to generate numerous trajectory hypotheses from short sequence segments and a trained classification model based on convolutional neural networks to verify the trajectory hypotheses. The detected targets are traced back from the verified trajectory. Furthermore, trajectory merge is conducted to link adjacent trajectory segments to detect more true targets to the fullest extent. We evaluate the proposed method on public datasets and compare it with existing methods. The experimental results indicate that the proposed method can detect small infrared aerial targets against cluttered backgrounds robustly, and it achieves better performance than existing methods. In addition, the proposed method can achieve real-time processing of sequence images.

The main contributions of this article are as follows.

1) A novel small aerial target detection method using trajectory hypothesis and verification is proposed in this article. The proposed method achieves better performance than do existing methods.
2) This study treats the trajectory verification as a classification problem. Moreover, a trained classification convolution neural network is adopted to verify the trajectory hypothesis. Because the network only uses trajectory information, it achieves superior generalization performance for different scenes.

The rest of the article is organized as follows: Section II summarizes the related works. In Section III, this article analyzes the trajectory characteristics of small infrared aerial targets. Details of the proposed method are presented in Section IV. Section V conducts extensive experiments and presents the experimental results. Section VI concludes the article.

II. Related Works

As mentioned above, the existing small target detection methods are mainly classified into single-frame-based methods and multiple-frames-based methods. In addition, we also summarize the most recent representative deep-learning-related methods separately.

A. Single-Frame-Based Methods

Single-frame-based methods detect small infrared targets using a single image. They mainly use the difference between the target and the background. Based on the characteristics used, such methods can be further subdivided as follows.

1) Target-Appearance-Characteristic-Based Methods: These methods detect small infrared targets mainly using the target's appearance. In remote imaging, a small infrared target is often modeled as a spot in an image. Its appearance distribution is determined by the imaging system. Kim and Lee [2] enhance target signals having a Gaussian distribution using a multiscale Laplacian of Gaussian (LoG) operator. Zhang et al. [3] adopt isotropic distributions to approximate the target. Although such methods are easy to implement and efficient, they are not suitable for detecting targets in cluttered backgrounds.

2) Background-Large-Spatial-Spread-Characteristic-Based Methods: Based on remote imaging and infrared imaging characteristics, researchers model the background approximately using common or uniform components. Compared with the background, a small target has a small spatial spread. A small target is detected by subtracting the estimated background. Gao et al. [4] use infrared patch image (IPI) reconstruction to model the background. Lu et al. [5] propose an effective online updating method to improve the traditional dictionary learning algorithm for infrared small target detection. With the aim of eliminating the negative effects of cluttered backgrounds, Zhou et al. [6] introduce the $l_{1/2}$ metric and double graph regularization in sparse component modeling. The robust principal component analysis (RPCA) decomposition is adopted to detect small infrared targets in [15]. Li et al. [16] regard small target detection as the difference between the original image and the background image, and propose a tensor-based minimum mean multiplication method. To solve the problem of detecting dim targets on high-brightness backgrounds, Han et al. [17] propose using a three-layer estimation window to detect targets at different scales. A new self-regularized weighted sparse (SRWS) model detection method is proposed in [18]. Song et al. [19] improve the propagation filter at the block level to estimate the background. Liu et al. [20] formulate an implicit regularizer by plugging a denoising neural network and using the weighted sum of the weighted tensor nuclear norm to achieve a more accurate background estimation. This type of method performs well when the background satisfies the large spatial spread assumption, but it cannot handle cluttered backgrounds satisfactorily. Furthermore, the background modeling is often time consuming.

3) Local-Characteristic-Difference-Based Methods: Small infrared targets and backgrounds belong to different components in images and exhibit different characteristics. The characteristic difference in the target-centered local region is used to detect a small infrared target. Zhao et al. [7] extract the contrast information of infrared small targets from the maximum tree and minimum tree. Li and Zhang [8] utilize the local steering kernel to estimate the inherent structure of the local image patch. The minimum contrast is taken as the output value after the contrast based on direction units is calculated in eight directions in [9]. Xiong et al. [21] propose a new method for detecting small infrared targets based on a local gradient field characteristic contrast measure (LGFCCM). Different improvements to the basic local contrast strategy are also utilized to detect small infrared targets in [22], [23], [24], and [25]. These methods suffer from a high false-alarm rate for cluttered backgrounds.

B. Multiple-Frames-Based Methods

Single-frame-based methods are easy to implement. However, they have trouble in detecting small infrared targets

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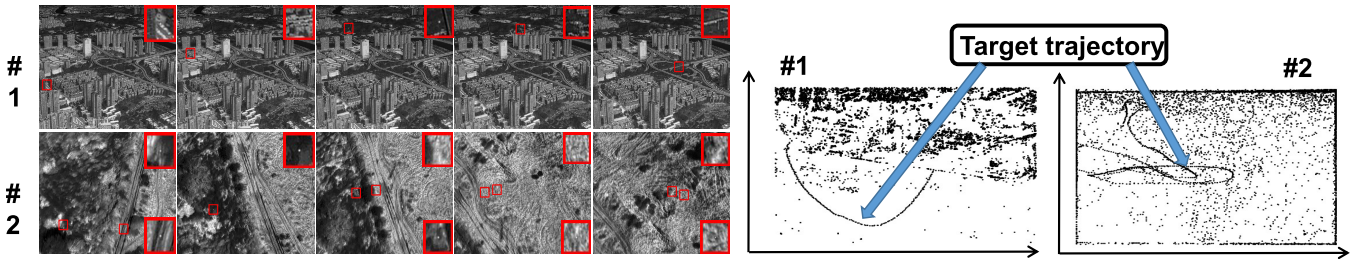


Fig. 1. Sample image sequences of small aerial infrared target. (Left) Sample images from sequence #1 and #2. Targets are labeled by rectangles. (Right) Remapped target candidates. The target candidates are remapped in the same coordinate system.

against cluttered backgrounds. Multiple-frames-based methods introduce temporal cues in target detection via association between successive frames. Sun et al. [26] detect the target in each frame through the pixel detection of interest and the LightGBM model trained, and they model the local smoothness and global continuity of the target trajectory as short-strict constraints and long-loose constraints. Kwan and Budavari [10] utilize optical flow technology to improve the detection performance of small infrared moving targets in low-quality and long-distance infrared video. A small target detection scheme based on spatiotemporal cues and multidirectional morphological filtering is proposed in [11]. Liu et al. [27] propose a low-rank approximation method by nonconvex tensor to accurately estimate clutter backgrounds. Zhang et al. [28] propose a new spatiotemporal tensor model to detect small and dim infrared targets considering edges and corners. Zhou et al. [29] combine the inherent tensor rank measure (ITRM) and spatial structure to improve the recovery of background components. Such methods cannot deal with the situation of rapid background changes. Moreover, the algorithm implementation is more complex, which makes it difficult to meet real-time requirements.

C. Deep-Learning-Relevant Methods

Traditional small infrared target detection methods rely on a variety of prior knowledge and assumptions. Such methods usually perform well only in specific scenes. Deep learning networks (DNNs) have achieved considerable success in many vision applications, e.g., detection, classification, and segmentation. Therefore, some researchers have tried to introduce supervised deep learning methods into the field of infrared small target detection to make use of the powerful feature learning ability of neural networks. To improve the problem that small targets disappear in the deep convolution map, Dai et al. [12] integrated the low-level features of small targets into high-level features at a deeper level and embed the prior knowledge about the local contrast of the target into the learning network. An efficient end-to-end network ISTNet is proposed in [13] by combining an image filtering module with a small target detection module. Du et al. [14] design effective small anchor points based on a shallow layer of ResNet50. Wang et al. [30] propose two context aggregation networks and connect them using U-skip connections. UNet is used in [31] as the backbone to maintain resolution and semantic information. Gupta et al. [32] use a lightweight neural network to enhance the traditional spatial-based detection processing chain. Wang et al. [33] propose obtaining a coarse target area and filter out the background using a region proposal network (RPN), and they use a transformer to model the attention between pixels in the coarse target area.

Most existing deep-learning-related work can be summarized as single-frame-based methods. Such methods mainly rely on the spatial information in infrared images but ignore the full utilization of temporal cues. Meanwhile, current methods based on deep learning require comprehensive coverage of training data, so the generalization performance of such methods is not good.

III. Analyzing Small Infrared Aerial Target Trajectory Characteristics

This study focuses on small aerial target detection. In remote imaging, the target is often presented as a single spot in the image. Due to the motion of the imaging platform, the cluttered background in the image sequence changes rapidly. Existing methods have difficulty in dealing with such cases. This study analyzes numerous small infrared aerial target image sequences. The movement of the aerial target follows the basic laws of physics. In general, targets in successive frames form continuous and smooth long trajectories. Fig. 1 presents two sample image sequences.

Sample sequences #1 and #2 contain 200 and 400 frames, respectively. The changes in the backgrounds are caused by the motion of the imaging system. This study adopts the IFDM to eliminate the background motion, extracts the candidate target from the single frame and maps the candidate targets to the same coordinate system. The remapped results in Fig. 1 show that the trajectories of the true targets are continuous and smooth, whereas the noise does not form continuous trajectories. To quantitatively verify the continuity and smoothness of the true target trajectory, this article evaluates the detection results of two sample sequences based on the trajectory constraint method in [26]. Table I presents the number of trajectories detected in sample sequences #1 and #2 under different trajectory lengths, as well as the precision rate (P) and recall rate (R) of the detection results. In the detection result for sample sequence #1, with the increase in trajectory length, the number of trajectories detected decreases constantly, while the precision rate increases constantly. Although the recall rate decreases, it maintains a high level. When the trajectory length threshold is set to 15, the method detects only a true trajectory, and all the false trajectories are eliminated. The success of the trajectory constraint method on sample sequence #1 shows that

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TABLE I
TRAJECTORY CANDIDATES AND DETECT RESULTS OF DIFFERENT LENGTHS WITHIN SAMPLE SEQUENCES IN [26] (P IS PRECISION RATE AND R IS RECALL RATE)

Length / Sequence	3	5	7	9	11	13	15
	number/(P, R)	number/(P, R)	number/(P, R)	number/(P, R)	number/(P, R)	number/(P, R)	number/(P, R)
#1	45/(0.15, 0.98)	28/(0.32, 0.94)	17/(0.46, 0.94)	6/(0.67, 0.91)	4/(0.75, 0.85)	3/(0.86, 0.81)	1/(0.99, 0.81)
#2	52/(0.26,0.93)	36/(0.33,0.90)	29/(0.45,0.88)	15/(0.54,0.86)	5/(0.57,0.65)	3/(0.76,0.54)	1/(0.99, 0.38)

TABLE II
PARAMETER SETTINGS OF THE TRADITIONAL TRAJECTORY CONSTRAINT METHOD IN [26]

Parameter	Setting
Link cost threshold σ_1	0.2
Velocity threshold σ_2	10
Trajectory segment similarity threshold σ_3	0.1
Trajectory segment stopped threshold σ_4	5
Length percentage μ	0.3
Time window length L	20

the true target trajectory does have smoothness and continuity, and a longer trajectory makes it easier to eliminate the false trajectories.

However, because of the strong assumptions about the trajectory, existing trajectory constraint methods cannot adapt to the complex trajectory or the poor continuity and smoothness caused by the interframe matching error. As shown in Fig. 1, compared with sequence #1, the target motion trajectories in sequence #2 are more complex and diverse, which results in poor detection results in sequence #2. In Table I, with the increase in the trajectory length, the number of detected trajectories in sequence #2 decreases, while the precision rate increases, which indicates that the number of false trajectories is decreasing. On the other hand, the recall rate of the detection results has plummets. The reason is that the approach in [26] cannot accurately model complex trajectories in sequence #2 with strong assumptions or effectively extend the true trajectories continuously. When the trajectory length increases, the true trajectory segment and the false trajectory segment are both deleted. Furthermore, as shown in Table II, existing methods that are based on trajectory constraints usually need to set numerous parameters to utilize various prior knowledge about the target trajectory. The setting of these parameters makes the algorithm unable to adapt to the complex and variable situation of a small target trajectory.

Traditional methods based on trajectory constraints cannot accurately model the trajectory of the target and have difficulty adapting to a complex trajectory and low trajectory smoothness. By contrast, the powerful feature extraction capability of convolutional neural networks has been fully practiced in the classification task. Therefore, this study explores the use of convolutional networks to select the true trajectory from a number of trajectory hypotheses by trajectory verification. Section IV presents the details of the proposed method.

IV. METHODOLOGY

In this study, trajectory verification is regarded as a classification problem for convolutional neural networks. Furthermore, a small aerial target detection method based on trajectory hypothesis and verification is proposed. The proposed method first extracts target candidates from each single image using the IFDM. Then, based on the continuity and smoothness characteristics of the target trajectory, a target trajectory hypothesis about the required length is generated by linking the target candidates from adjacent frames. Finally, the trajectory hypothesis is sent to a trained, classification convolutional neural network for verification. The detected target is traced back from the verified trajectory. Furthermore, trajectory merging is performed to link adjacent trajectory segments. The overall architecture of the proposed method is shown in Fig. 2.

A. Target Candidate Detection in Each Frame

The image sequence captured by the camera has the characteristic of continuity. The background has a similar consistency across two or three consecutive frames. Therefore, it can be considered that the difference between different images is mainly caused by the movement of the target.

This study performs interframe difference operations on three consecutive images in time, subtracts the corresponding pixel points of different frames, and judges the absolute value of the gray difference. If the absolute value exceeds the set threshold, it is considered to be caused by moving objects in the image and these areas are labeled as foreground pixels. The remaining areas of the image are treated as background pixels.

This study uses the mean and standard deviation of a single image to threshold the difference image, and the threshold is calculated as follows:

$$t_{\text{th}} = \text{mean}(d_\text{img}) + k * \text{std}(d_\text{img}) \tag{1}$$

where mean and std represent the mean and standard deviation of the difference image, respectively, and k is a constant that is determined empirically.

The IMU is a piece of standard equipment in modern airplanes. With appropriate settings, the IMU device can directly output the relative pose between adjacent frames. And, the time-consuming process of image registration based on IMU is almost negligible.

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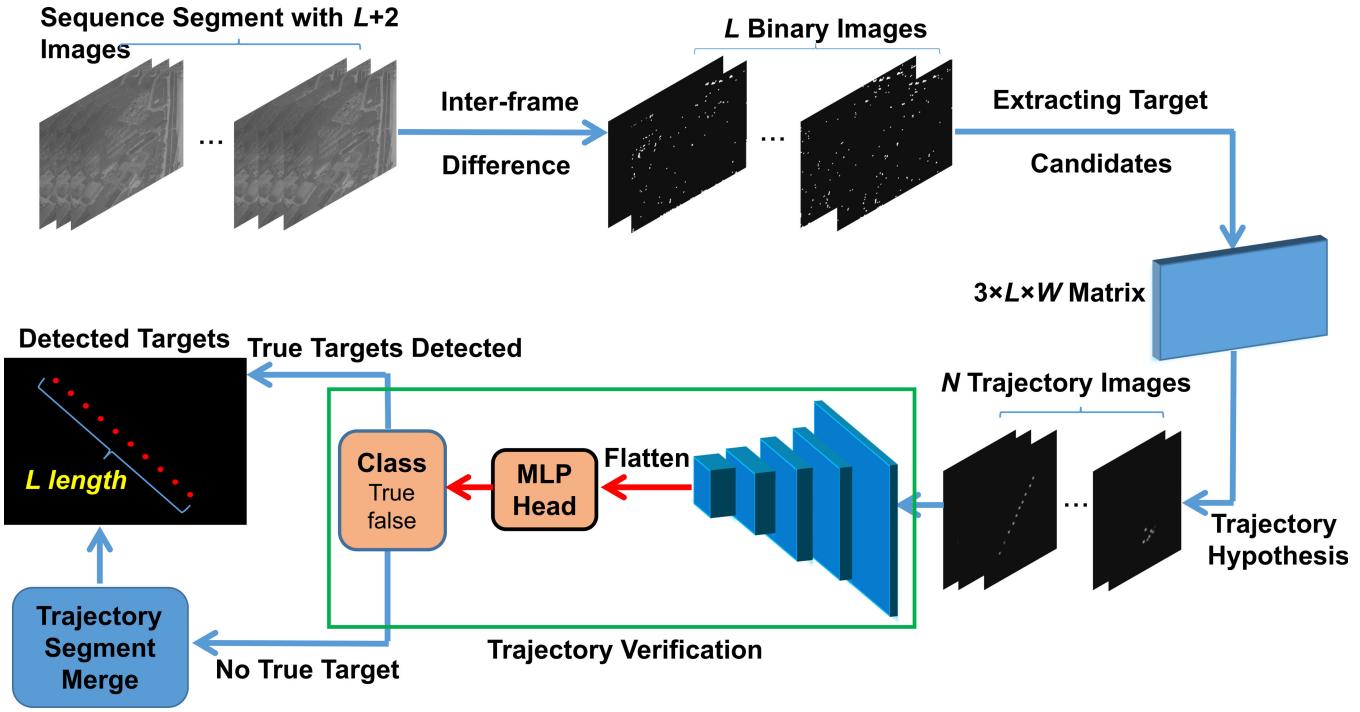


Fig. 2. Overall architecture of our method. The proposed method mainly consists of three parts: target candidates' detection from each frame, trajectory hypothesis, and trajectory verification.

The output data of IMU are the relative rotation ΔR and relative translation ΔT between adjacent IMU. The relative rotation R_{CI} and relative translation T_{CI} between IMU and camera can be obtained through the imaging platform. So it can be obtained that the relative rotation ΔR^c and relative translation ΔT^c between adjacent cameras are as follows:

$$\Delta R^c = R_{\mathrm{CI}} \Delta R R_{\mathrm{CI}}^T \tag{2}$$

$$\Delta T^c = -R_{\mathrm{CI}} \Delta R R_{\mathrm{CI}}^T T_{\mathrm{CI}} + R_{\mathrm{CI}} \Delta T + T_{\mathrm{CI}}. \tag{3}$$

The homography matrix between adjacent frames can be calculated by

$$H = K\left(\Delta R^c - \frac{\Delta T^c n^T}{d}\right) K^{-1} \tag{4}$$

where K represents camera intrinsic matrix, n^T and d represent the normal vector and intercept of the plane P where the image point is located, respectively. In this article, IMU equipment is used to measure the continuous interframe motion in short time with high precision. Since the time interval between adjacent frames is short, the drift of IMU device can be basically ignored. When IMU data is lacking, the SURF [34] or ORB [35] +RANSAC method can be used to calculate the homography matrix between adjacent images.

To generate the target trajectory hypotheses conveniently and quickly for sequence segments, this study sorts the targets segmented, according to the size of the target area from high to low, and takes the first W targets in each single frame. When the number of targets in a single-frame image is less than W, invalid data are added to ensure that there are W candidate targets in each frame. Then, a multiframe candidate target matrix of size $3 \times L \times W$ over a length of L consecutive frames is generated. The three channels store the x coordinate, y coordinate, and gray value of the target center point in the candidate target matrix, respectively.

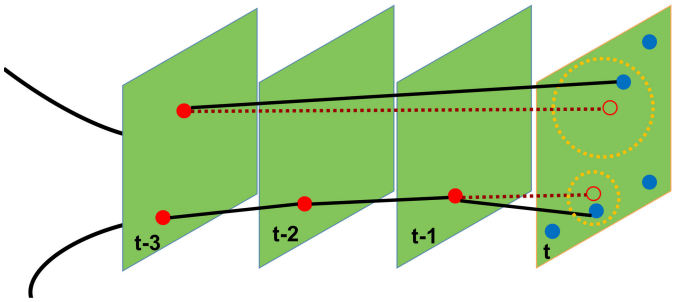


Fig. 3. THG based on continuity. Targets on existing trajectories are represented by red circles and target candidates on the current frame are represented by blue circles.

B. Trajectory Hypothesis Generation (THG)

After obtaining the candidate target matrix as mentioned above, the continuity and smoothness of trajectories is used to generate the target trajectory hypotheses from the target matrix for verification. As shown in Section III, the movement forms of small targets vary and their trajectories are complex and diverse. However, their trajectories are all continuous. This study generates trajectory hypotheses from target candidates according to the distance constraint criterion. Its principles are as follows.

1) As shown in Fig. 3, the candidate targets in the current frame are linked with the existing trajectories according to the distance constraint. When the last target of the trajectory segment belongs to the previous frame, we search for qualified candidate targets in the current frame, that is, candidate targets whose distance from the

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end of the trajectory segment is within the specified range. Then, the trajectory segment is linked with the qualified candidate targets to generate new trajectories.

2) Considering the fact that some frames have no true targets segmented, this study allows for a maximum of two frames missing between adjacent targets in existing trajectories, as shown in the upper part of Fig. 3. Moreover, the distance constraint between the two should be proportionally expanded according to the number of frames between them.
3) After all trajectory hypotheses are extracted according to the above principles, we set the proportionality coefficient of trajectory length c to delete the trajectories whose number of candidate targets is less than $c \times L$, which is convenient for the training and verification of the classification network.

The details of THG are presented in Algorithm 1.

Algorithm 1 THG

Input: Candidate target matrix $\mathbf{M} \in \mathbb{R}^{3\times L\times W}$
Output: Trajectory set $\mathbf{T} = \{T^k\}_N$
$\mathbf{T} = \{\}, i = 0$
while $i \leq L-1$ **do**
  if $i = 0$ **then**
    $\mathbf{T} \leftarrow [\mathbf{T}, \mathbf{M}(:, i, :)]$
  else
    $\mathbf{A_1} \in \mathbb{R}^{2\times len(\mathbf{T})\times 1} \leftarrow$the end coordinates of $\mathbf{T}$
    $\mathbf{A_2} = \mathbf{M}(0:2, i, :) \in \mathbb{R}^{2\times 1\times W}$
    $\mathbf{A_1'} \leftarrow repeat(\mathbf{A_1}, W, axis = 2)$
    $\mathbf{A_2'} \leftarrow repeat(\mathbf{A_2}, len(\mathbf{T}), axis = 1)$
    $\mathbf{D} = abs(\mathbf{A_1'} - \mathbf{A_2'})$
    $\mathbf{D} \in \mathbb{R}^{len(\mathbf{T})\times W} \leftarrow sqrt(\sum_{j=0}^{1} \mathbf{D}(j, :, :)^2)$
    // $index(T^k)$ is the end frame number of the kth trajectory
    for $t = 0 : W$ do
      flag $= 0$
      for $k = 0 : len(\mathbf{T})$ do
        if $\mathbf{D}(k, t) \leq \sigma_1 \times (i - index(T^k))$ **and** $(i - index(T^k)) \leq \sigma_2$
          $T^k \leftarrow [T^k, A_2(:, :, t)]$
          flag $= 1$
      if flag$== 0$ **then**
        $\mathbf{T} \leftarrow [\mathbf{T}, \mathbf{M}(:, i, t)]$
  $i = i + 1$
for $k = len(\mathbf{T}) : 0$ do
  if $len(T^k) \leq c \times L$ **then**
    delete T^k

C. Trajectory Hypothesis Verification (THV)

This study maps the trajectory hypothesis generated above to a mask image with the same size as the input image. Because of the small size of the hypothesized trajectory, we crop the hypothesized trajectory according to the minimum bounding rectangle and resize the cropped area to the size of input image. Then, multiple trajectory mask images are generated and sent to the classification network for verification to detect the true trajectory segments. As shown in Fig. 4, the trajectory formed by true targets approximates a smooth curve, while the trajectory generated by false targets is chaotic and cannot be fit using curves. The geometric feature distributions of true trajectories and false trajectories differ greatly, so we can distinguish them using convolutional neural networks.

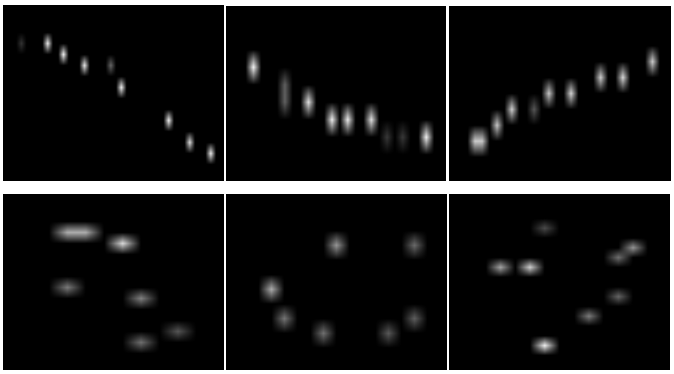


Fig. 4. Sample trajectory mask images for classification. (First row) True target trajectory mask images. (Second row) False target trajectory mask images.

Note that the imaging platform for the synthetic dataset used in this study has smooth motion. To speed up the target detection, the proposed method does not transform targets along a trajectory segment to the same coordinate system. However, in other cases, the swing amplitude of the camera and imaging platform may be large over a short period of time, so it is necessary to transform the target to the same coordinate system to maintain the continuity and smoothness of the trajectory.

D. Trajectory Segment Merge (TSM)

In Section IV-C, the trajectory hypotheses are sent to the classification network to pick out true trajectories. However, the partial sequence segments may be a situation where no hypothesized trajectory is generated that meets the required length, or where the true trajectories are not classified, which reduces the recall rate of the proposed method. Furthermore, some trajectory hypotheses generated via linking true targets are discarded in the previous step because of insufficient length. To improve the recall rate of the proposed method, it is necessary to try to redetect the trajectories in some sequence segments where the true targets are not detected.

As described in [26], the characteristics of adjacent trajectory segments that correspond to the same target are as follows: 1) the trajectory segments do not overlap in time; 2) the trajectory segments extend close to each other; and 3) the speed values of different trajectory segments are close to each other. Inspired by this, the true trajectory segments detected in the previous sequence segment are used for a similarity merging calculation with the hypothesized trajectories of the current sequence segment without sufficient length. This study takes two trajectory segments, T^1 and T^2, as samples to explain the definition of the similarity measure, as shown in Fig. 5. T^1 is the true trajectory segment detected in the previous sequence segment, and T^2 is the trajectory hypothesis discarded in the current sequence segment due to insufficient

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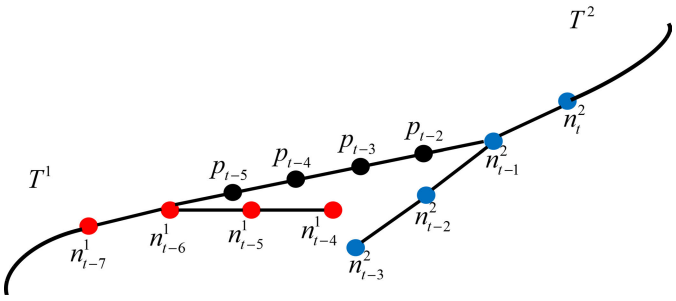


Fig. 5. Similarity measure for trajectory segment merging.

length. Note that the positions of the missing target on T^1 and T^2 are supplemented based on the assumption of uniform motion over a short time.

T^1 and T^2 are adjacent in time. T^1 ends at frame $(t-4)$ and T^2 starts from frame $(t-3)$. If T^1 and T^2 belong to the same target trajectory, this study links n^1_{t-6} and n^2_{t-1}. Under the uniform motion constraint, the interpolated target positions are $\{p_{t-5}, p_{t-4}, p_{t-3}, p_{t-2}\}$, as shown in Fig. 5. The distances between the interpolated target positions and the actual positions are used to define the similarity measure $s(T^1, T^2)$ between T^1 and T^2, as in the following equation:

$$s(T^1, T^2) = \frac{(t-1)-(t-6)-1}{\sum_{k=t-5}^{t-4} \|n^1_k - p_k\|_2 + \sum_{k=t-3}^{t-2} \|n^2_k - p_k\|_2}. \quad (5)$$

The equation above is a detailed description of the similarity definition of two trajectory segments, T^1 and T^2. The row in similarity matrix $S = [s(T^i, T^j)]_{N\times M}$ is the true trajectory segment of the previous sequence segment, and the column is the chosen trajectory hypotheses of the current sequence segment. The binary link matrix A is given in the following equation:

$$A(i,j) = \begin{cases} 1, & S(i,j) \geq \sigma_3 \\ 0, & S(i,j) < \sigma_3 \end{cases} \quad (6)$$

where σ_3 is a threshold set for the degree of relaxation of in the similarity of adjacent trajectory segments. In $S = [s(T^i, T^j)]_{N\times M}$, any trajectory segment T^i can be merged with only one T^j. Therefore, we select the trajectory hypothesis that corresponds to the column with the largest similar measurement value in each row for the merge.

V. Experiments

A. Experimental Settings

1) Baseline Methods: In this section, we compare the proposed method with twelve state-of-the-art methods, including the single-frame-based methods (IPI [4], improved local contrast measure (ILCM) [36], multiscale local contrast measure (MLCM) [37], absolute average difference weighted by cumulative directional derivatives (AADCDD) [38], reweighted infrared patch-tensor (RIPT) model [39], attentional local contrast network (ALCNet) [12], dense nested attention network (DNANet) [40], non-negative IPI model based on partial sum minimization of singular values (NIPPS) [41]), multiple-frames-based method (using LightGBM and trajectory constraints (ULTC) [26], TIPI [42], nonconvex tensor low-rank approximation (NTLA) [27], and multiple subspace learning and spatial–temporal IPT (MSLSTIPT) [43]). Among these comparison methods, ALCNet and DNANet are also detection methods based on deep learning. For the compared methods, most of the parameters are the default settings released by the author, and a few have been modified to fit the dataset used in this article. Detailed parameter settings of the comparison method are shown in Table III. Note that some of the parameters of ULTC have previously been presented in Table II.

TABLE III
Parameter Settings of Baseline Methods

Method	Hyper-parameters settings
IPI	Patch size=50, sliding step=20, $\lambda = 1/\sqrt{max(m,n)}$, $\eta = 0.99$
ULTC	Local region size=$\{3, 7, 11, 15\}$
ILCM	Window's size=8, step size=4
MLCM	$K = 4$, the size of target region=$\{3, 5, 7, 9\}$
NIPPS	Path size=50, sliding size=10, $\lambda = \frac{L}{min(m,n)}$, $L = 2$, $r = 5 \times 10^{-3}$
AADCDD	Internal window size=$\{3, 5, 7, 9\}$, external window size =19
RIPT	Patch size=50, sliding step=20, $\lambda = L/\sqrt{min(I, J, P)}$, $L = 1$, $h = 10$, $\epsilon = 0.01$, $\varepsilon = 10^{-7}$
TIPI	Patch size=50, patch step=5, patch lenth=3
NTLA	$\lambda_{tv} = 0.005$, $\delta = 0.5$, $k = 9$, $v_{min} = 0.85 \times 255$, $\lambda_s = (H/((max(m,n) \times L)^{1/2}))$
MSLSTIPT	Path size=30, $L = 6$, $p = 0.8$, $k = 9$, $v_{min} = 216$, $\lambda = 1/\sqrt{n_3 max(n_1, n_2)}$

2) Parameters Setting: For the proposed method, the parameter setting are listed in Table IV. As for the classification network used in the proposed method, considering that the trajectories are mostly simple curves without too many features, ResNet-50 [44] is selected as the backbone network for extracting features. To satisfy the needs of the binary classification tasks, the fully connected layer of ResNet-50 is replaced with the concatenation of two fully connected layers whose output dimensions are 256 and 2, respectively. A cross entropy loss function is used for binary classification tasks. For optimization, we adopt stochastic gradient descent (SGD) [45] as an optimizer with an initial learning rate of 0.1 and the strategy described by He et al. [46] for the weight initialization, the StepLR learning rate reduction strategy with a step_size = 20, a total of 100 epochs, a momentum of 0.5, and a batch size of 8.

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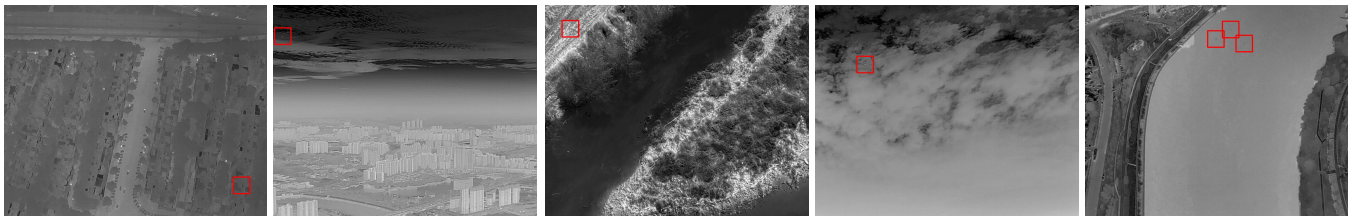


Fig. 6. Sample images picked form the dataset expand-SIATD. (Left to right) Building, ground-sky, vegetation, sky, water. The targets are labeled by rectangle.

TABLE IV
PARAMETER SETTINGS OF THE PROPOSED METHOD

Parameters	Setting
Sequence segment's length L	10
Target candidates number of each frame W	100
Distance constraint threshold σ_1	5
Cross frame length threshold σ_2	3
Trajectory segment similarity threshold σ_3	1/15
Trajectory length proportionality coefficient c	0.7

All experiments are conducted using python 3.7 and Pytorch on a computer with a 3.2-GHz AMD Core R7-5800H CPU, RTX 3060 GPU, and 16-GB RAM. To speed up the detection, the classification model trained based on Pytorch is converted into TensorRT model for inference.

3) Dataset: The SIATD dataset was presented in [26]. According to the Society of Photo-Optical Instrumentation Engineers (SPIE), all objects with an area less than 80 pixels are called small targets. Therefore, based on the SIATD dataset, we expand the maximum size of small targets to 9×9 and build a new dataset: expand-SIATD. The expand-SIATD dataset will be released later.

As with the SIATD dataset, the expand-SIATD dataset covers five scenes, including buildings, ground-sky, vegetation, sky, and water. Furthermore, the expand-SIATD dataset is classified into training set and test set, which contain 175 image sequences, respectively. We show some example images in Fig. 6.

We train ALCNet, DNANet, and the proposed method on the training set of the expand-SIATD dataset and test all methods on the test set of the expand-SIATD dataset.

4) Evaluation Metric: Due to remote imaging, infrared small targets usually appear as dots in an image. As shown in Fig. 6, the edges of small targets are blurry, so it is not meaningful to accurately segment the shape of the target in practice. We think that accurate calculation of the center position of small targets can play a better role than segmenting the target area in practical tasks. Furthermore, the F-measure (F) metric, which is widely used in target detection task, is used to evaluate our work. This metric combines the precision rate (P) and recall rate (R) of the evaluated method. It is defined in the following equation:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}$$
$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$$
$$F = \frac{(1+\beta^2)P * R}{\beta^2 P + R} \tag{7}$$

where TP, FP, and FN denote the number of correctly detected targets, number of wrongly detected targets, and the number of missed targets, respectively. We set $\beta^2 = 1$ in this study.

We define the target as correctly detected if the location detected is within the 3-pixel neighborhood of the ground truth.

B. Ablation Study

To analyze the effectiveness of different components of the proposed method, an ablation study is conducted on the test set of expand-SIATD. Table V classifies the proposed method into five parts: IFDM, THG, THV, and TSM. Then, the THG, THV, and TSM are analyzed according to the order in the proposed method.

1) Effectiveness of THG: The experimental results are listed in Table V. Using IFDM+THG, the precision rate on the whole test set reaches 0.45, indicating that nearly half of the false targets are screened out by utilizing the continuity and smoothness of the trajectory.

2) Effectiveness of THV: It can be seen from the data in the third row of Table V that the THV method is further used to classify trajectory hypotheses. The comprehensive evaluation of the F score indicator is further increased and the precision rate of target detection is greatly improved. The precision rate in three scenes reaches more than 90%, and the precision rate in all scenes reaches more than 88%. Except for the vegetation scenario, the reduction of recall rate in the other four scenarios is controlled at about 10%. In the partial sequences of the vegetation scene, due to the large and irregular jitter of the camera, the true trajectories lack smoothness, so they are difficult to classify correctly.

3) Effectiveness of TSM: It seems that the front components attempted to improve the precision rate of small target detection while ignoring the recall rate. Therefore, as shown in the last row of Table V, we add the TSM step to improve the recall rate of target detection.

4) Time-Consuming Analysis: The accumulated time consumed by the proposed method is shown in the last column of Table V. It can be seen that in the proposed method, the average consumption time of each frame is about 29 ms. Among them, the IFDM step takes about 6 ms on average, the THG step takes about 5 ms to generate trajectory hypotheses, the classification step THV takes about 15 ms to calculate, and the TSM step TSM takes an average of 3 ms/frame. The

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TABLE V

ABLATION STUDY OF DIFFERENT COMPONENTS OF THE PROPOSED METHOD. THE EXPERIMENT IS CONDUCTED ON THE TEST SET OF EXPAND-SIATD WITH $F1$-MEASURE. THE EXPERIMENT COMPARES THE PERFORMANCE OF DIFFERENT COMPONENTS UNDER FIVE SCENARIOS AND OVERALL DATA, AND SHOWS THE TIME CONSUMPTION OF DIFFERENT COMPONENTS. THE BEST PERFORMANCE IS IN BOLDFACE

Method	building	ground-sky	vegetation	sky	water	whole	Time/s
	$F(P, R)$	$F(P, R)$	$F(P, R)$	$F(P, R)$	$F(P, R)$	$F(P, R)$	
IFDM	0.12(0.07, 0.94)	0.29(0.21, 0.97)	0.13(0.08, 0.91)	0.30(0.21, 0.98)	0.16(0.09, 0.96)	0.20(0.13, 0.96)	0.006
THG	0.34(0.26,0.79)	0.64(0.59,0.81)	0.46(0.39,0.67)	0.76(0.70,0.85)	0.38(0.29,0.77)	0.52(0.45,0.78)	0.011
THV	0.76(0.88,0.70)	0.79(0.97,0.70)	0.64(0.89,0.51)	0.86(0.98,0.78)	0.79(0.90,0.72)	0.77(0.92,0.68)	0.026
TSM	**0.77**(0.84, 0.73)	**0.80**(0.96, 0.72)	**0.65**(0.81, 0.55)	**0.87**(0.96, 0.81)	**0.81**(0.88, 0.76)	**0.78**(0.89, 0.71)	0.029

time required to calculate interframe motion using IMU data is almost negligible, and the vast majority of the time consumed in IFDM steps is used for candidate target extraction.

C. Comparison With State of the Art

The results in Fig. 7 show that the baseline algorithms have difficulty detecting infrared small aerial targets under cluttered backgrounds. Clutter backgrounds and point targets introduce considerable difficulty for IPI and ILCM. As shown in Fig. 7, the above methods barely detect true targets in five scenarios. TIPI has trouble with rapid changes in modeled backgrounds. Many background components remain in the target image, as shown Fig. 7. In contrast, MLCM, NIPPS, AADCDD, RIPT, NTLA, and MSLSTIPT can detect more true targets, but these methods also introduce many false alarms, especially with cluttered backgrounds, e.g., buildings and vegetation. ULTC fails to detect some true targets on cluttered backgrounds. The performances of ALCNet and DNANet are better than most traditional comparison methods, but as shown in Fig. 7, there are still some scenarios in which the true target is not detected. The proposed method can correctly detect the targets in all the scenarios in Fig. 7, and its performance is better than that of methods.

The quantitative evaluation of each method on the test set is reported in Table VI, including the results of each scene type and the entire test set. Note that the precision rate, recall rate, and F-measure shown in Table VI are the average values of different sequences.

The precision rate, recall rate, and F-measure of the twelve methods on all test sets are shown in Table VI. As Table VI shows, the cluttered background presents considerable challenges to existing single-frame-based methods. IPI attempts to separate background and target based on the low rank of the background and the sparsity of the target. This method performed only slightly better against a sky background. The reason is that more complex scenes of vegetation, buildings, ground-sky, and water do not fit the assumption of a simple background. ILCM performs poorly in all scenarios. MLCM introduces a multiscale local contrast mechanism, and its comprehensive performance in the five scenes is much better than that of ILCM, especially in simple scenes such as ground-sky. However, it is difficult for the local comparison mechanism to detect small targets with low SNR and targets in cluttered backgrounds, so the overall recall rate of MLCM is low, and the detection performance is poor on cluttered backgrounds such as buildings and vegetation. AADCDD is also based on a local comparison mechanism. This method has a high recall rate on a simple sky background, but it is also not suitable for cluttered backgrounds. RIPT has the highest recall rate among the five scenarios, and its F-measure in sky and water scenarios is also superior to that of the proposed method. However, the precision rate of RIPT in complex environments such as vegetation and buildings is extremely low. Similar to RIPT, NIPPS has a high recall rate on simple backgrounds, e.g., sky and water, but has a low-precision rate. ALCNet and DNANet are deep-learning-based approaches. ALCNet combines the local comparison mechanism of small targets with a neural network to construct a model-driven deep learning, small target detection method. Its comprehensive performance is better than that of most comparison methods. However, compared with the proposed method, the precision rate and recall rate of ALCNet are mediocre. DNANet has a perfect precision rate in all scenarios; however, the main problem with DNANet is its relatively low recall rate. Like ALCNet, DNANet is not good at detecting small targets that have a low SNR. Although DNANet's comprehensive index F-measure is slightly lower than that of the proposed method, its detection performance is actually very close to that of the proposed method.

Compared with the single-frame-based methods, the multiframe-based methods introduce additional temporal cues. TIPI is improved based on IPI, which treats small targets as special sparse noise components of complex background noise and uses temporal cues to detect targets. However, as shown in Table VI, the performance of TIPI in all scenarios is not ideal, indicating that this assumption is difficult to apply in complex contexts. Furthermore, TIPI is too time consuming to be applied in practice. The short-strict and long-loose constraints enable ULTC to achieve a high-precision rate, but the strong assumptions about trajectory also lead to a low recall rate. Both NTLA and MSLSTIPT are detection methods based on background estimation. The two methods perform well on simple backgrounds such as sky and water surfaces, but they perform poorly on complex ground scenes due to low-rank assumptions about the background.

The proposed method achieves the highest F-measure on ground-sky scenes, and it has a higher average F-measure than other comparison methods on the overall test set. However, the detection performance of the proposed method in the vegetation scene is relatively poor. There are two main reasons:

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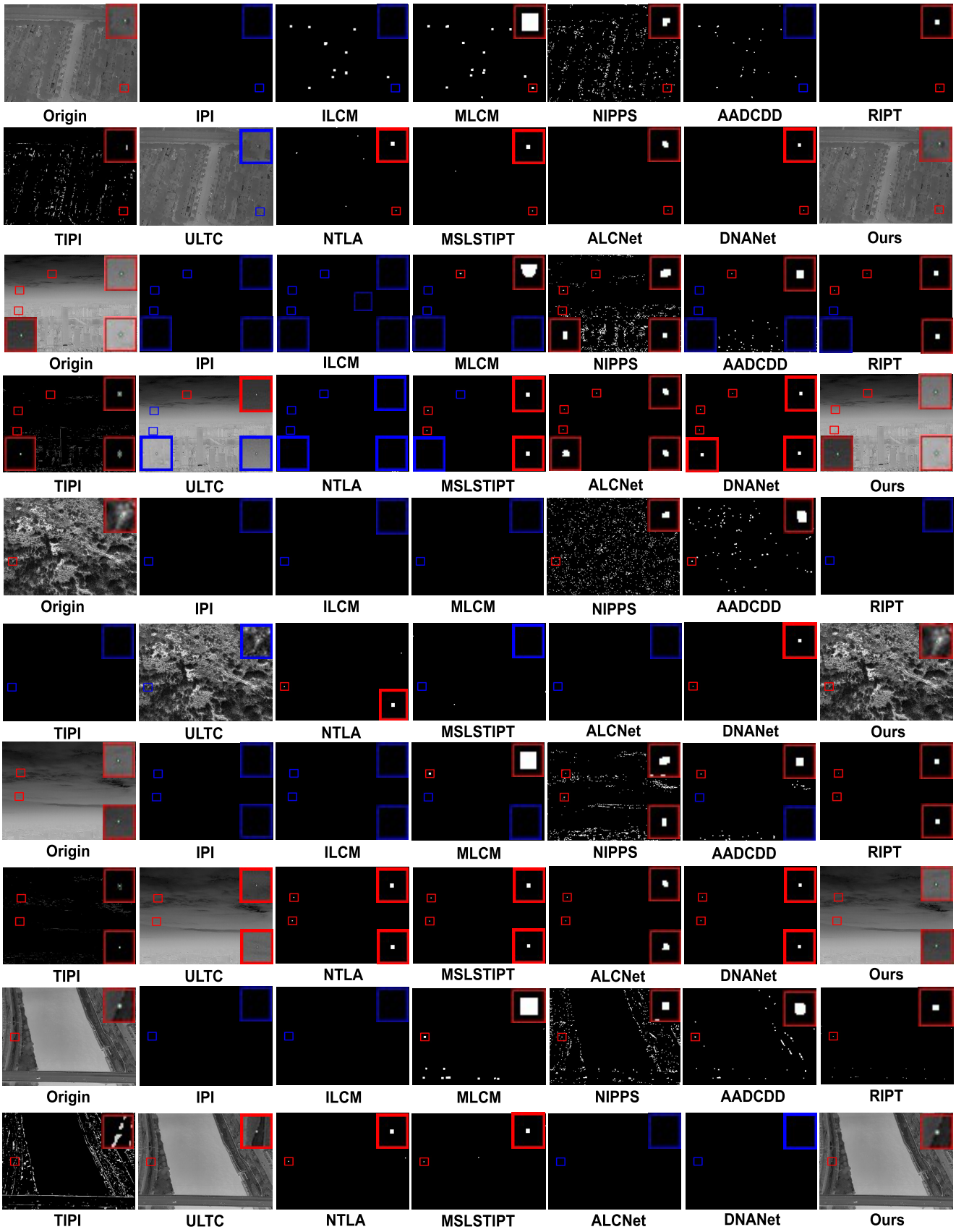


Fig. 7. Detection results of the proposed method and baseline methods on expand SIATD. Correct detection results are highlighted in red boxes, missing targets are highlighted in blue boxes.

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TABLE VI
QUANTITAVE EVALUATION OF IPI, ILCM, MLCM, NIPPS, AADCDD, RIPT, TIPI, ULTC, NTLA, MSLSTIPT, ALCNET, DNANET, AND OURS ON THE TEST SET OF EXPAND-SIATD (P FOR PRECISION RATE, R FOR RECALL RATE, AND F FOR F-MEASURE)

Method	building	ground-sky	vegetation	sky	water	whole	Time/s
	$F(P, R)$	$F(P, R)$	$F(P, R)$	$F(P, R)$	$F(P, R)$	$F(P, R)$	
IPI	0.03(0.02, 0.06)	0.01(0.01, 0.01)	0.00(0.00, 0.00)	0.21(0.44, 0.17)	0.01(0.02, 0.01)	0.05(0.10, 0.05)	4.93
ILCM	0.00(0.00,0.00)	0.01(0.02,0.01)	0.00(0.00,0.00)	0.02(0.01,0.03)	0.00(0.00,0.00)	0.01(0.01,0.01)	0.05
MLCM	0.17(0.43, 0.22)	0.35(0.91, 0.23)	0.37(0.54, 0.33)	0.83(0.98, 0.74)	0.34(0.28, 0.59)	0.41(0.63, 0.42)	0.11
NIPPS	0.00(0.00, 0.12)	0.02(0.01, 0.40)	0.02(0.01, 0.05)	0.06(0.03, 0.96)	0.02(0.01, 0.94)	0.02(0.01, 0.50)	180
AADCDD	0.02(0.01, 0.21)	0.05(0.03, 0.40)	0.02(0.01, 0.30)	0.29(0.25, 0.68)	0.04(0.02, 0.39)	0.08(0.06, 0.40)	0.11
RIPT	0.18(0.10, **1.00**)	0.53(0.37, **1.00**)	0.03(0.01, **0.95**)	**0.98**(0.98, **0.99**)	**0.86**(0.80, **0.99**)	0.52(0.45, **0.99**)	129
TIPI	0.02(0.01, 0.03)	0.12(0.10, 0.12)	0.06(0.04, 0.07)	0.19(0.15, 0.20)	0.14(0.13, 0.17)	0.11(0.09, 0.12)	2571
ULTC	**0.78**(0.95, 0.66)	0.74(0.76, 0.65)	0.75(0.99, 0.60)	0.79(0.93, 0.69)	0.72(0.98, 0.57)	0.75(0.92, 0.63)	0.12
NTLA	0.54(0.44, 0.84)	0.58(0.71, 0.50)	0.50(0.45, 0.72)	0.84(**1.00**, 0.77)	0.83(0.83, 0.86)	0.63(0.60, 0.79)	10.20
MSLSTIPT	0.67(0.55, 0.90)	0.71(0.90, 0.60)	0.58(0.50, 0.78)	0.87(0.98, 0.81)	0.77(0.72, 0.86)	0.73(0.71, 0.83)	25.35
ALCNet	0.28(0.27, 0.60)	0.74(0.96, 0.61)	0.65(0.78, 0.61)	0.75(**1.00**, 0.61)	0.72(0.89, 0.62)	0.63(0.78, 0.61)	1.00
DNANet	0.76(**0.99**, 0.62)	0.76(**0.99**, 0.63)	**0.75**(**0.99**, 0.62)	0.76(**1.00**, 0.62)	0.78(**1.00**, 0.65)	0.76(**0.99**, 0.63)	0.11
Ours	0.77(0.84, 0.73)	**0.80**(0.96, 0.72)	0.65(0.81, 0.55)	0.87(0.96, 0.81)	0.81(0.88, 0.76)	**0.78**(0.89, 0.71)	0.029

1) due to the large swinging and floating of the camera in vegetation scenes, part of the image is relatively fuzzy. At the same time, due to the large distance between adjacent targets, no effective trajectory can be generated and 2) the large swing of the camera over a short period of time causes some target trajectories to lack smoothness and not be correctly classified.

The last column of Table VI also shows the detection time of the baseline methods and the proposed method on a single-frame image. As shown in Table VI, ILCM, MLCM, and AADCDD, three methods based on single-frame detection, have faster detection speed but poor detection effect. The detection process of IPI, NTLA, MSLSTIPT, NIPPS, RIPT, and TIPI requires iteration, so it is time-consuming, and the processing time of TIPI on a single frame reaches approximately 42 min, which seriously limits the application of the algorithm in practice. The time-consuming ULTC is mainly used to extract target candidates in a single frame. ALCNet processes an image per second on average, and the computing speed has yet to be improved. The speed of DNANet is fast and the detection result is good. Using IMU data and optimizing model, the proposed method achieves the minimum time consumption among all methods. And, the processing speed of the proposed method is about 34 frames/s, which can realize real-time processing.

The performance of DNANet is similar to that of the proposed method on the expand-SIATD dataset. Because generalization performance is an important performance indicator for network models, during the experiments on the expand SIATD dataset, we save the detection models generated by DNANet and the proposed method. Then, the generalization performance of the two models on other datasets is compared. A real small target dataset called real dataset [47] is used in a generalization performance comparison experiment in this article. The image size of real dataset is 256 × 256, where the target size is between 1 × 1 and 4 × 5. Its background is mainly a ground vegetation environment, which is relatively complex, and the background of a few image sequences is a sky scene. We select seven sequences, called sequences 1–7, from real dataset for the generalization experiment, and these seven sequences contain a total of 6696 images.

Because the motion of the imaging platform of real dataset is not smooth, the proposed method transforms the targets in the trajectory segment to the same coordinate system, and then sends them to the classification network for verification. The test results are shown in Table VII and Fig. 8. It can be found that DNANet generates numerous false alarms, and its recall rate is lower than that of the proposed method. The proposed method has a relatively high-precision rate and recall rate, and

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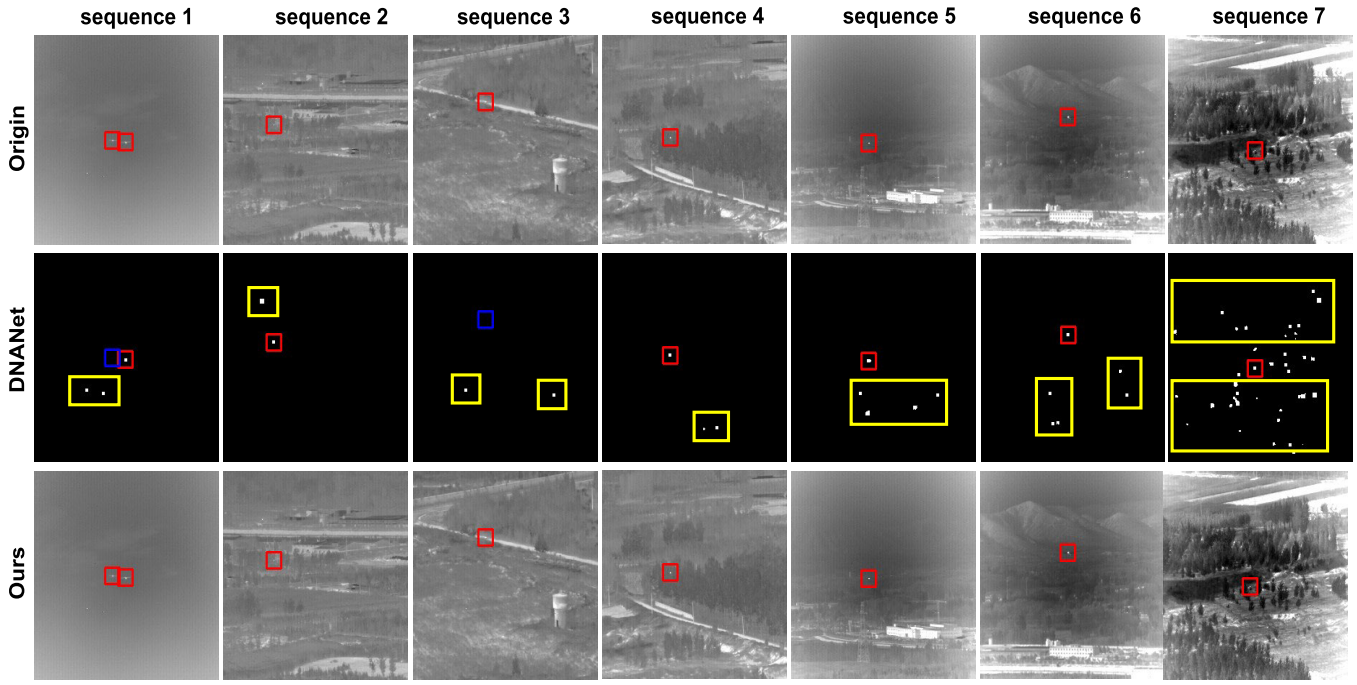


Fig. 8. Detection results of the proposed method and DNANet on seven sequences belonging to real dataset. Correct detection results are highlighted in red boxes, missing targets are highlighted in blue boxes, and false alarms are highlighted in yellow boxes.

TABLE VII

QUANTITATIVE EVALUATION OF DNANET AND OURS ON THE SEVEN SEQUENCES OF REAL DATASET (P FOR PRECISION RATE, R FOR RECALL RATE, F FOR F-MEASURE)

Sequence	DNANet	Ours
	$F(P, R)$	$F(P, R)$
sequence 1	0.27(0.24, 0.32)	0.72(0.94, 0.59)
sequence 2	0.38(0.28, 0.60)	0.74(0.93, 0.62)
sequence 3	0.38(0.29, 0.58)	0.77(0.89, 0.68)
sequence 4	0.33(0.33, 0.34)	0.83(0.86, 0.82)
sequence 5	0.19(0.11, 0.65)	0.62(0.66, 0.59)
sequence 6	0.21(0.13, 0.63)	0.77(0.85, 0.72)
sequence 7	0.05(0.02, 0.79)	0.88(0.83, 0.94)
whole	0.26(0.20, 0.56)	**0.76**(**0.85**, **0.71**)

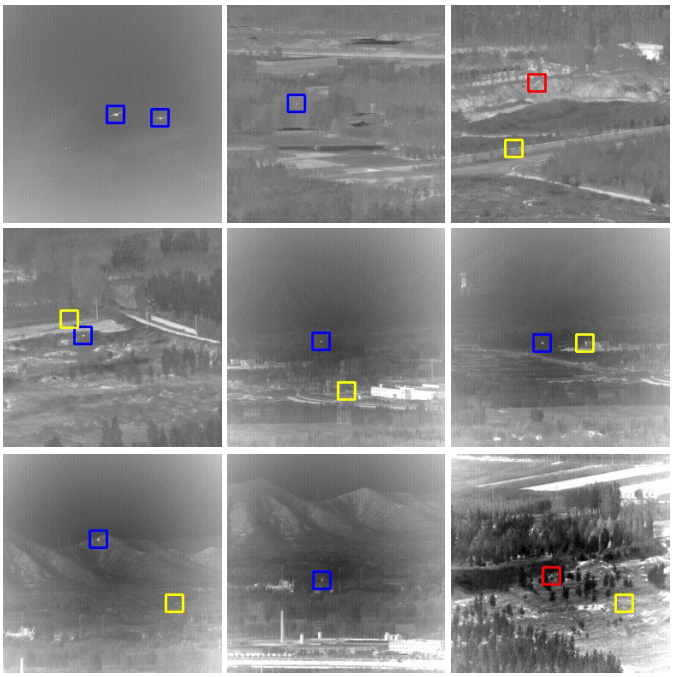


Fig. 9. Failure cases. Correct detection results are highlighted in red boxes, missing targets are highlighted in blue boxes, and false alarms are highlighted in yellow boxes.

its score on the comprehensive performance index F-measure is also much higher than that of DNANet, which indicates that the proposed method had better generalization performance on real dataset. DNANet learns the spatial characteristics of small targets simulated in the expand-SIATD dataset through a large amount of data. The shape and gray-level distribution of real targets are subject to many factors, whereas the simulation targets have an ideal Gaussian distribution. In addition, the background of images in real dataset is also obviously different from the expand-SIATD dataset. These factors cause the DNANet model trained on expand SIATD to perform poorly in the generalization experiment. We attribute the relatively good generalization performance of the proposed method to the learning of target trajectory characteristics. The background of the images in different datasets varies greatly, but the method proposed sends the trajectory graph composed of the extracted target trajectory into the network for verification, which is independent of the background, and the trajectory of the target in different datasets usually has similar smoothness and continuity.

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Although the proposed method shows better generalization performance, it also has limitations. As shown in Fig. 9, there are cases where true targets are not detected, and false alarms are generated in some images. This is mainly due to the large number of candidate targets extracted in a single frame, which means the hypothesized trajectories generated by many false targets in the sequence segment may also have smoothness. At the same time, the classification network based on ResNet-50 may have classification errors, leading to some true trajectories being wrongly judged as false trajectories. We will consider establishing a more powerful and accurate classification network in the future to improve the performance of the proposed method.

VI. Conclusion

We point out that the trajectories of small aerial targets have smoothness and continuity in short sequence segments. Based on this, an infrared small target detection method based on trajectory hypothesis and verification is proposed in this article. The proposed method uses IMU data to calculate the motion between adjacent frames. The IFDM is used to extract target candidates from a single frame. In short sequence segments, numerous trajectory hypotheses are generated from target candidates by simple distance constraints based on trajectory continuity assumptions. These trajectory hypotheses are then sent to the trained classification network for verification to detect the true trajectory segments. In this study, the similarity of adjacent trajectory segments is used to merge the trajectory segments to improve the recall rate of target detection. Experiments on public datasets show that the proposed method can detect small infrared aerial targets robustly and achieves better performance than existing methods. Moreover, using IMU data and optimizing classification model, the proposed method can achieve real-time processing of sequence images.

For future work, we seek to integrate single-frame candidate target extraction, trajectory hypothesis, and trajectory verification into a deep learning algorithm to build an end-to-end small target detection neural network.

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